

Understanding signal extraction from the 2021 10% sample

HPS Gaussian-process background analysis

24 September 2026

Version 6.3.6

The result. Moving the Gaussian signal template toward the reconstructed signal-MC core improves the fitted response. The Gaussian still represents only part of the asymmetric signal distribution. To make statements about the *full selected signal yield*, the study therefore calibrates the fit statistic using simulated experiments with a known injected signal yield.

This note explains that procedure and the evidence supporting it. It is a presentation revision of v6.3.5: the main-body numerical ensembles, fitted results and observed scans are unchanged. Appendices A–C add v16 TC and UC signal-MC core-shift studies. The signal-MC catalogue and core-shift comparison from v6.1 are included so that the choice of Gaussian center can be understood here without consulting another note.

Why the distinction matters. There are three separate questions. Where is the reconstructed signal core? How much does the fitted Gaussian yield change when signal is added? Which injected yields are compatible with the observed fit result? A shifted Gaussian addresses the first question, an injection study measures the second, and toy-calibrated tests answer the third. Moving the Gaussian alone does not solve all three.

For signal-MC injections on the GP mean background source at $A = 3s_0$ (where s_0 is the yield-error scale from background-only pilot fits), the added-signal response ranges from 0.208–0.422 for the central mass Gaussian and 0.265–0.833 for the shifted Gaussian across 60–240 MeV. A response of 0.5 means that adding 1,000 selected signal events increases the fitted Gaussian yield by about 500 events. It is not a detector efficiency. When the injected signal is itself a matched Gaussian, the shifted fit returns 0.963–0.980 of the added yield.

What is usable now. The finite-grid toy calibration provides a defined, testable procedure under the specified background sources and signal-MC shape. At $A = 5s_0$, independent evaluation experiments retain the true injected yield in 81–96% of cases for the GP mean source and 84–93% for the functional form source. Each fraction uses 100 experiments. These are conditional validation results with finite statistical precision, rather than a guarantee of exactly 90% coverage at every mass or for an unknown background shape.

Reading guide. Sections 1–2 explain the signal shape and fit. Sections 3–7 distinguish background-only bias, signal response and yield calibration. Section 8 explains the upper-limit construction. Sections 9–11 show the observed 2021 and combined results. Section 12 checks the cost of changing the fit window. Section 13 records the scope and reproducible sources.

The main search and calibration domain is 60–240 MeV. The 260 MeV signal-MC sample is shown only as a shape comparison. The report does not establish a global discovery probability or a detector-qualified physical exclusion.

1. Why move the Gaussian center?

A narrow generated resonance need not reconstruct as a symmetric Gaussian centered at its generated mass. Detector response and selection change the reconstructed distribution. In the supplied target-constrained (TC) signal Monte Carlo, the fitted core is displaced below the generated signal mass and a substantial tail extends toward higher reconstructed masses.

We compare two Gaussian extraction templates. The **central mass Gaussian** is centered at the tested signal mass m . The **shifted Gaussian** is centered at the empirical signal-core position $c(m)$:

$$c(m) = m - 3.2243308693 - 2.2139928114 \ln(m/150), \quad 60 \leq m \leq 240 \text{ MeV}.$$

Here m and $c(m)$ are in MeV; $m/150$ means $m/(150 \text{ MeV})$. This is a logarithmic relation for the *core displacement*, not a logarithmic probability model for the entire signal distribution.

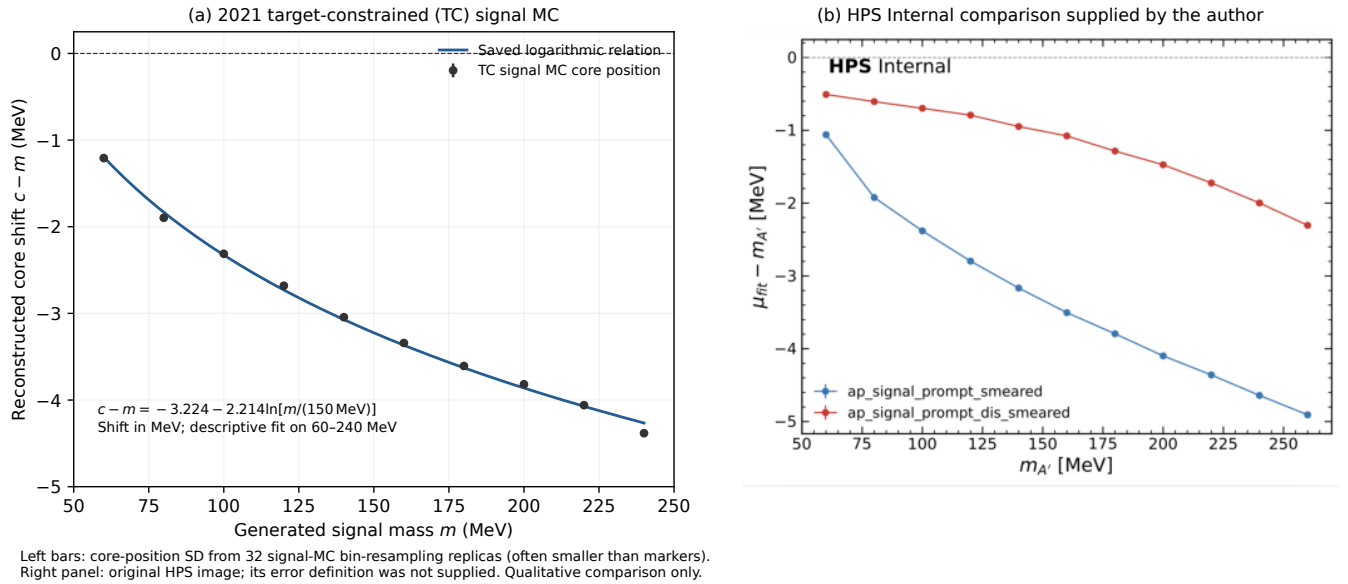


Figure 1. How to read the figure. Left: the v6.1 TC signal-MC core displacement and the logarithmic relation used in this study. Left error bars are the standard deviation of the fitted center over 32 signal-MC bin-resampling replicas, often smaller than the markers; they are not the 100-toy mean errors used later. Right: the HPS Internal image supplied with this revision, reproduced unchanged. Its vertical coordinate is fitted Gaussian mean minus generated mass; negative values indicate a lower reconstructed center. The blue and red curves are the two sample categories named in that image. Its error definition and reconstruction/selection equivalence to the present input were not supplied, so the right panel provides qualitative context, not a combined fit. The logarithmic curve on the left is the one used here.

Only the Gaussian center changes between the main fit choices. Both retain the same **reference analysis resolution**,

$$\sigma_{\text{ref}} = 0.00184825 - 0.001375m + 0.085875m^2, \quad m \text{ and } \sigma_{\text{ref}} \text{ in GeV}.$$

The signal-fit window and the region excluded from background training are both centered on that Gaussian and extend to $\pm 2.25\sigma_{\text{ref}}$. The fitted signal-MC core width used to compare shapes on the next pages is a different quantity.

1. Signal-MC catalogue: examine each sample first

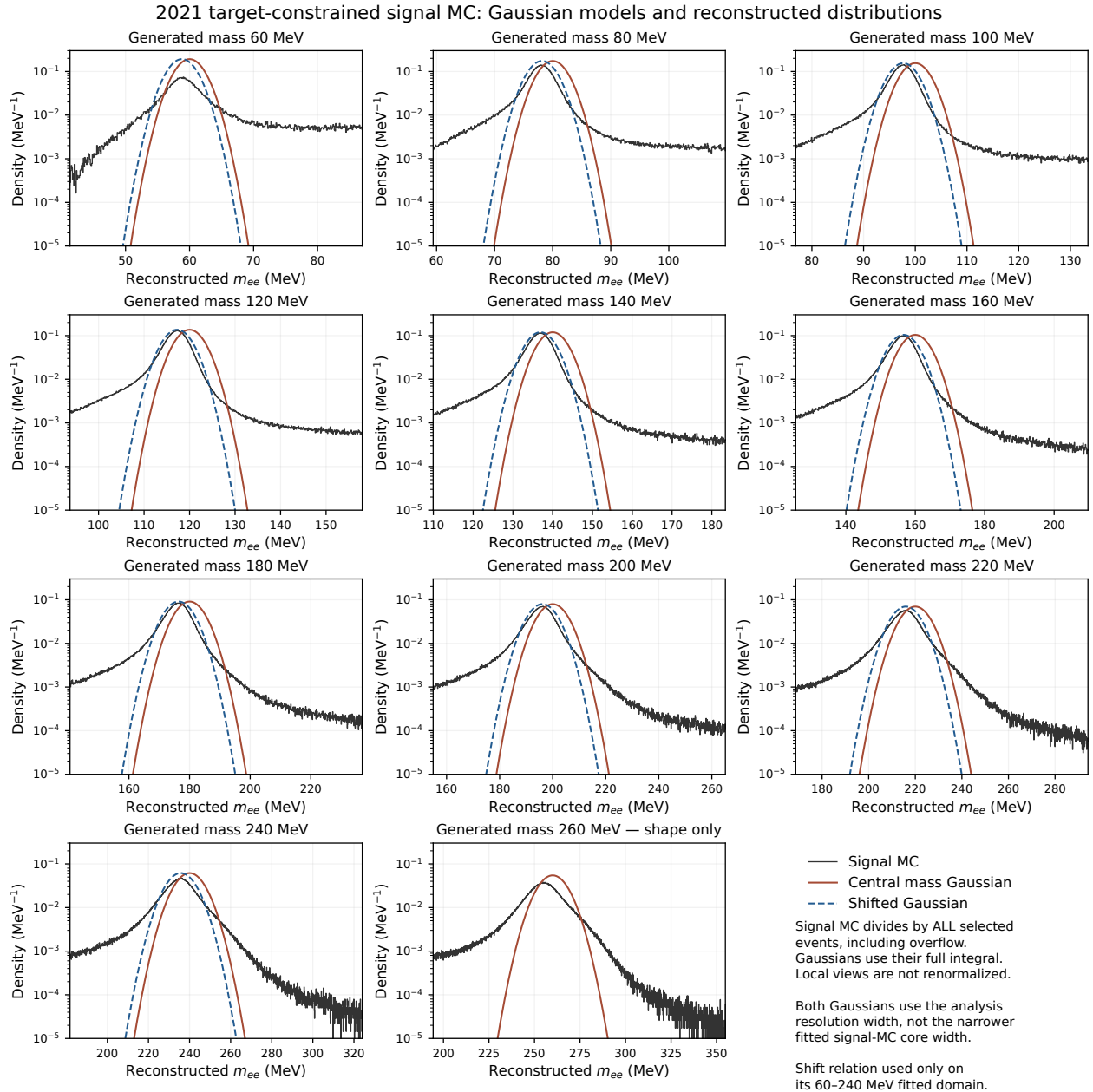
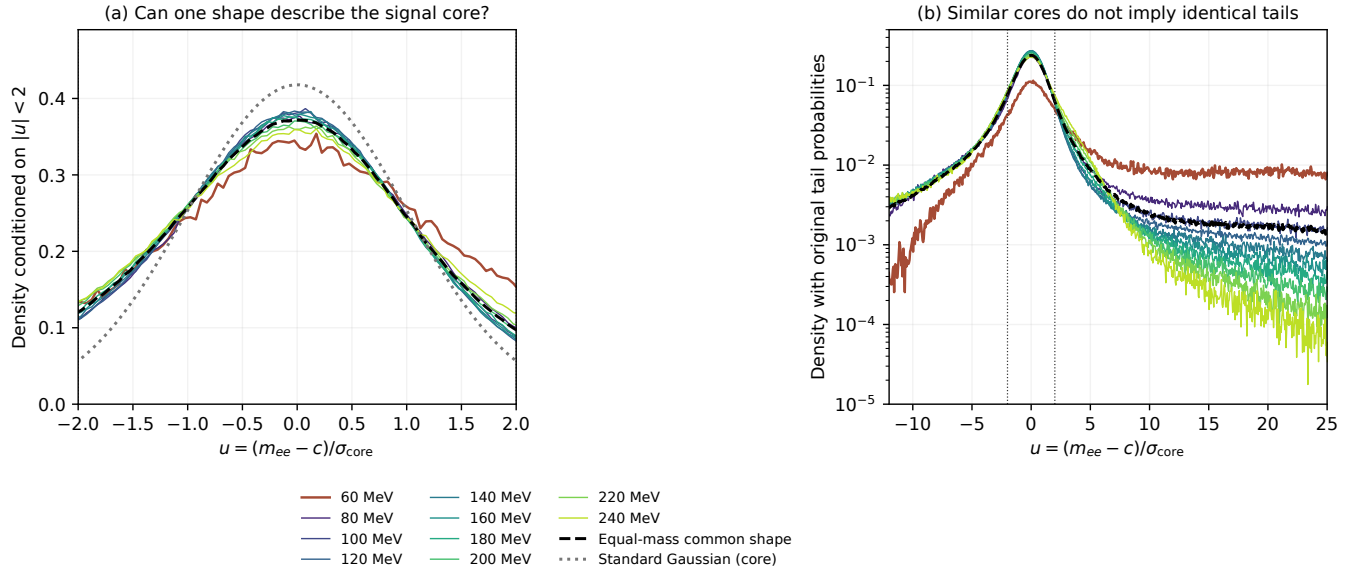


Figure 2. How to read the figure. Each panel shows one generated signal-MC sample with the central mass Gaussian and the shifted Gaussian used for extraction. Both Gaussians have the reference analysis width, not the fitted signal-MC core width. The logarithmic vertical scale exposes tails over several orders of magnitude. These are signal-MC probability densities, not background toys; their full normalization is retained when only a local mass range is shown. The 260 MeV sample is a shape control outside the 60–240 MeV calibration domain and has no logarithmically shifted curve. No new fit was performed for this revision.

The catalogue makes two features visible together: the central peaks move relative to their generated masses, and the selected signal is not fully Gaussian. An extraction template can locate the core well while still returning less than the full selected signal yield.

1. Can one shape describe the signal cores?



c and σ_{core} : fitted signal-MC core centre and width. Right: probabilities divide by all selected events, including overflow. No toy bars.

Figure 3. How to read the figure. The signal-MC distributions are aligned by their individual fitted core centers and widths. The horizontal coordinate is $u = (m_{rec} - c_{core})/\sigma_{core}$, so a unit of u is one fitted core width. The comparison tests whether the central shapes look similar after alignment; it does not imply that their full tails or acceptances are identical. Left: each curve is normalized within $|u| < 2$ to compare core shapes. Right: the original core and tail probabilities are retained. The dashed black common shape is the equal-mass average of the ten aligned empirical distributions, not an analytic Gaussian fit. The gray dotted reference is a standard Gaussian normalized within $|u| < 2$. These curves reuse the saved v6.1 histograms and core fits; no toy error bars are shown.

Answer. A common empirical shape describes the cores reasonably well from about 80 MeV upward. Its peak is approximately Gaussian, with heavier shoulders than a standard Gaussian. The tails remain visibly different, and the low-mass sample needs separate treatment. Thus a common core description is a useful guide to the Gaussian center, but is insufficient to calibrate the total signal yield.

The low-mass qualification. The supplied independent signal-MC samples are generated at 60, 80, 100, ..., 260 MeV. There is no independently simulated 65 MeV sample in this catalogue. The fitted core of the 60 MeV sample is about 58.8 MeV. The acceptance-loss concern near 65 MeV is therefore illustrated here by the actual 60 MeV input, not by relabeling a sample. Its distorted distribution is consistent with the low-mass geometric acceptance turn-on discussed in the signal study; this plot does not isolate the cause. These selected distributions alone do not measure a separate acceptance value at 65 MeV.

The saved v6.1 diagnostic puts 32.39% of its stored-range probability within two fitted core widths, compared with 60.73–71.47% for 80–240 MeV. The figure also retains the tiny overflow probability (below 0.016% in these samples). This helps explain why the 60 MeV response is particularly small in the injection study: fitting its central region with a Gaussian cannot account for the large fraction of selected signal outside that region. Subsequent sections measure that effect directly rather than assuming it away.

2. What the Gaussian fit measures

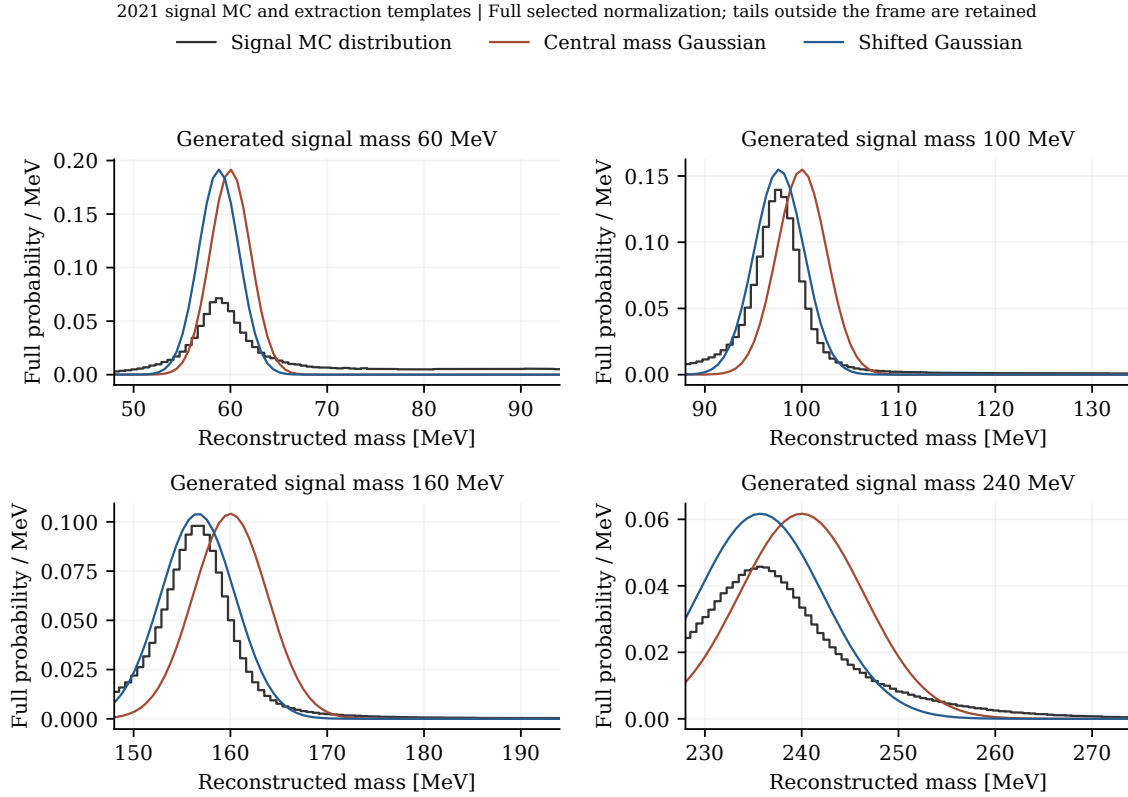


Figure 4. How to read the figure. Black: the selected signal-MC distribution used to inject events. Red: the central mass Gaussian. Blue: the shifted Gaussian, also called the *log core Gaussian* because its center follows the logarithmic law in Section 1. Both Gaussians use the same reference analysis resolution. All curves retain their full probability normalization; the displayed mass range omits some tails without renormalizing what remains. These are the same extraction prescriptions as in the catalogue; the linear vertical scale emphasizes the core mismatch.

Let A be the expected number of **full selected signal events**. This counts the whole selected signal distribution, including events outside the analysis support and fit window. Let $\hat{A}$ be the signed Gaussian yield returned by the fit, and $\hat{\sigma}$ its reported standard error. Negative fitted yields are allowed so that background fluctuations are treated symmetrically; expected bin counts must remain positive.

The background is estimated with Gaussian-process regression (GPR) from the sidebands, meaning the bins outside the excluded signal region. Each toy or observed spectrum supplies its own sideband counts. The GP kernel settings remain fixed, but its predicted background mean and covariance are recomputed for that spectrum. The likelihood uses Poisson bin counts and correlated background nuisance parameters; the error $\hat{\sigma}$ comes from the curvature of the fitted profile likelihood.

Two effects reduce $\hat{A}$ relative to the full signal yield. First, the Gaussian fits only a projection of an asymmetric signal distribution. Second, signal in the training sidebands can be absorbed into the estimated background. With the background known exactly, the separate window study still finds response 0.378–0.857. The corresponding GP response divided by that control is 0.703–0.974. Moving the center helps, but neither effect is removed automatically.

2. How the simulated experiments are organized

A **toy experiment** is a simulated binned count spectrum. Background counts are drawn independently in each bin from Poisson distributions with fixed expected counts. We use two background sources:

GP mean background source. The expected bin counts are the pinned full-support GP arithmetic mean. We fluctuate counts around that fixed model; we do not draw a new GP background function for each toy.

Functional form background source. The expected bin counts come from the archived anchored `fSigPowExpQ` function. This alternative tests sensitivity to one specified change in background shape. Its earlier local assessment does not establish that it describes every part of the data globally.

Signal events are added according to the fixed selected signal-MC probabilities, including training sidebands and outside-support categories. At fixed expected signal yield A , their total fluctuates according to a Poisson distribution. These are *not* injections with an exactly fixed event total. Neither the signal MC nor the Gaussian is renormalized inside the fit window.

Three separate sets of experiments. First, 100 GP-source background-only *pilot* experiments define $s_0(m)$, the mean reported yield error from the shifted Gaussian. This fixes the signal-strength unit $A = z s_0$. The label z therefore specifies injected yield in pilot-error units; it is not a discovery significance.

Next, each source has 100 new *calibration* backgrounds. These determine the background-only bias, the response, and the reference distributions used by the yield tests. Those choices are frozen. Finally, each source has another 100 *independent evaluation* backgrounds to test the frozen calibration. These evaluation experiments were called “heldout” in earlier versions: they were simply not used to determine the calibration.

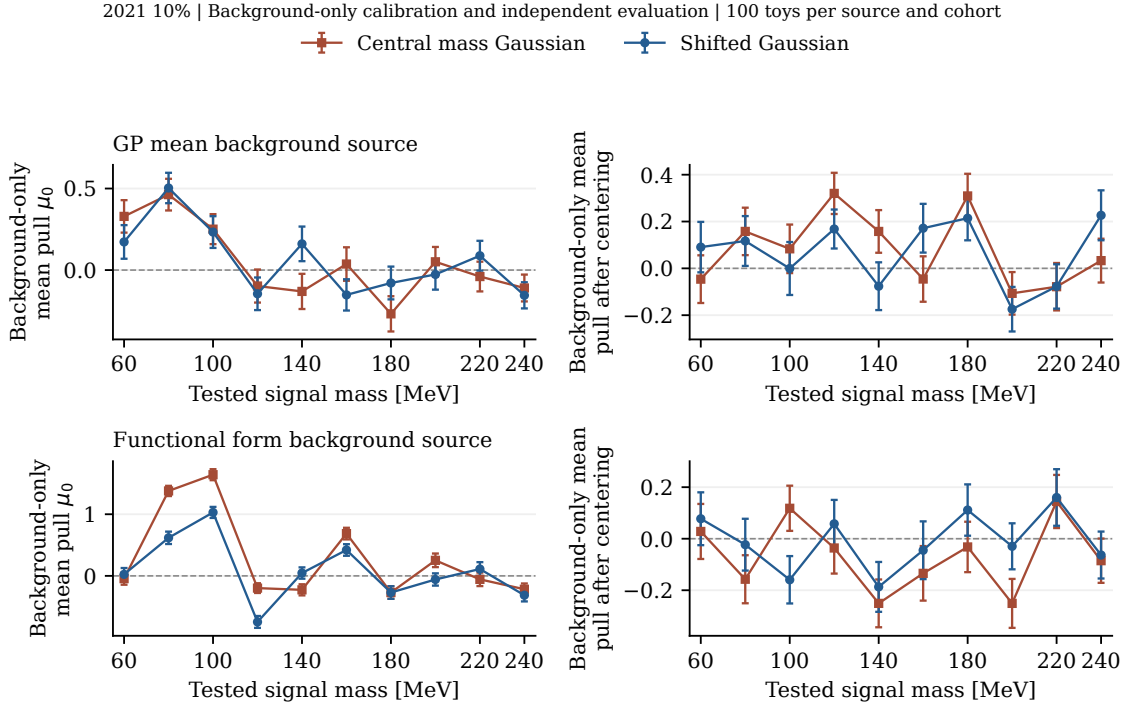
Calibration uses $z = 0, 1, 2, 3, 4, 5, 6, 8, 10, 12, 16, 20, 24$; evaluation uses $z = 0, 1, 3, 5$. Gaussian signal controls use $z = 1, 3, 5$ on the GP mean source. The same expected yield is used for both Gaussian centers and both background sources.

Within a source and cohort, the same background experiment is reused across masses, strengths and fit choices. The two Gaussian fits also receive identical signal counts. Such comparisons are paired: their differences remove common fluctuations. Different sources and cohorts have independent streams. There are 500 main background experiments, with many fits to each, not 76,000 independent experiments.

Main calculation	Attempted	Valid
Pilot fits	2,000	2,000
Calibration fits	52,000	52,000
Evaluation fits / truth profiles	22,000 / 22,000	22,000 / 22,000
Gaussian asymptotic limit endpoints	16,000	16,000

Uncertainty. Means have sample-standard-error bars: the sample standard deviation divided by $\sqrt{100}$. Bootstrap calculations resample whole toy experiments 2,000 times, keeping masses, injected strengths and analysis choices together; the two background sources use separate streams. Widths have bootstrap uncertainty. Fractions retain all 100 trials and use exact two-sided 95% Clopper–Pearson intervals. The caption specifies which of these bars appears here.

3. Measure background-only bias before adding signal



Means: sample standard error (sample SD / $\sqrt{100}$). Widths: bootstrap standard error.
 Bootstrap: 2,000 whole-toy resamples keep masses, strengths and Gaussian choices together; sources use separate streams.
 Binomial fractions: all 100 trials retained; exact two-sided 95% Clopper-Pearson intervals.
 Mean bars after calibration condition on the frozen calibration table; its uncertainty is assessed separately.

Figure 5. How to read the figure. Both columns are background only ($A = 0$). Left: calibration mean pull $\mu_0 = \widehat{A}_0/\widehat{\sigma}_0$. Right: mean of $\widehat{A}_0/\widehat{\sigma}_0 - \mu_0$ in a separate evaluation cohort. Top: GP mean background source; bottom: functional form background source. Bars are sample standard errors of means from 100 experiments. The right-hand bars condition on the frozen left-hand calibration estimate. The uncertainty conventions are also printed on the figure.

A background-only fit can return a nonzero average signal yield even though no signal was injected. We call this the **background-only fitted-yield bias**, $\delta = \widehat{A}_0$, measured in events. The **background-only mean pull**, $\mu_0 = \widehat{A}_0/\widehat{\sigma}_0$, instead measures displacement in units of the fit's own error. They are different averages and are not interchangeable.

Subtracting the frozen μ_0 centers a pull diagnostic. Subtracting δ corrects an additive yield bias. Neither operation restores a missing fraction of injected signal. Nor does subtracting a mean change the width of a distribution.

For the shifted Gaussian, independent background-only mean pulls after centering range from -0.174 to 0.227 for the GP mean source and -0.187 to 0.160 for the functional form source. The corresponding uncentered pull widths are 0.832 – 1.131 and 0.893 – 1.121 . Finite calibration and evaluation samples leave visible fluctuations; zero mean and unit width are targets to check, not assumptions to impose.

4. Signal response before and after calibration

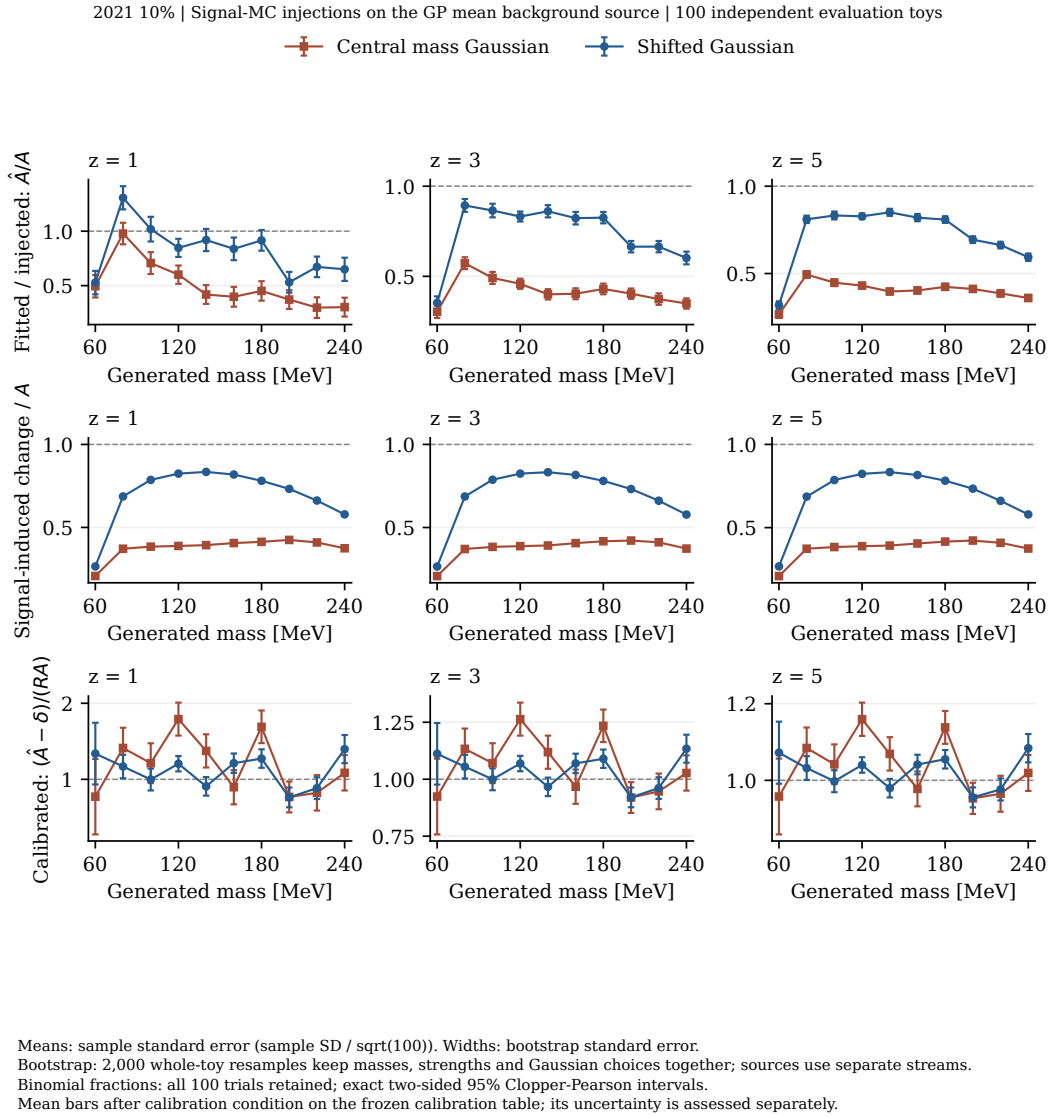


Figure 6. How to read the figure. Signal-MC injections on the GP mean background source. Columns use $A = zs_0$ with $z = 1, 3, 5$. Top: fitted-to-injected yield ratio $\hat{A}/A$. Middle: change caused by adding signal, $(\hat{A}_{s+b} - \hat{A}_b)/A$, with the same background experiment in both fits. Bottom: calibrated response $\bar{A}_c/\bar{A}$, where $A_c = (\hat{A} - \delta)/R$. The independent calibration cohort supplies δ and the response R measured at $z = 3$. A value of one means that the estimator returns the injected full selected yield on average. Each point averages 100 independent evaluation experiments. Bars show sample standard errors, conditional on the frozen calibration table; uncertainty from estimating that table is assessed separately.

The middle row isolates response to the added signal; the top row also contains background-only yield bias. The bottom row answers the requested calibration question directly. It applies the same frozen correction at all three strengths and tests it on new experiments. Its bars describe the evaluation mean for that frozen correction; they do not claim exact response under a different background source or signal shape.

A fixed subtraction of δ cancels in the middle-row difference, so it cannot repair a response below one. Dividing by R addresses that scale difference. Section 7 explains the uncertainty of the resulting yield estimator; Section 8 uses full toy distributions for inference instead of relying on this simple correction.

5. Repeat the test with a functional form background

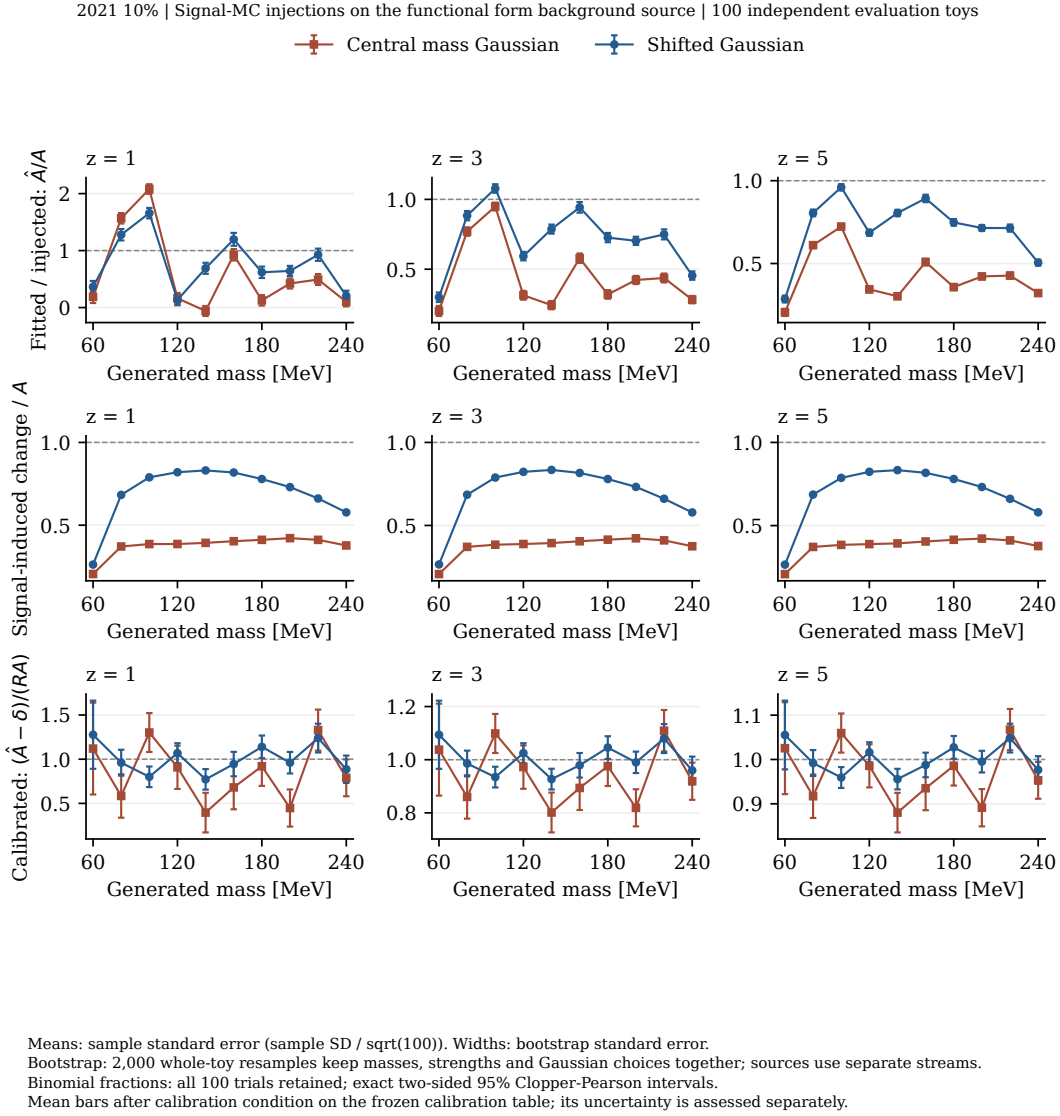


Figure 7. How to read the figure. Signal-MC injections on the functional form background source. The rows and strengths have the same definitions as in Section 4: fitted-to-injected yield, signal-induced change divided by injected yield, and calibrated response $(\hat{A} - \delta)/(RA)$. Here the correction is derived from an independent calibration cohort of the same functional form source. Expected signal yields match the GP-source study, while source ensembles are independent. Each point averages 100 independent evaluation experiments. Bars show sample standard errors, conditional on the frozen calibration table; uncertainty from estimating that table is assessed separately.

At $z = 3$, the shifted-Gaussian signal-induced response is 0.265–0.834 for this source, close to the GP-source range. The fitted-to-injected ratio can differ more because the background-only biases differ. Agreement of the incremental response under two chosen sources is useful evidence, but does not test every possible background shape.

As a separate check, injected Gaussians centered on the logarithmic law give shifted-fit response 0.963–0.980 on the GP mean source. This is much closer to one than the signal-MC response. Signal-shape mismatch therefore matters even after the center is corrected. Gaussian controls have independent signal draws and receive no signal-MC response correction.

6. What a pull says about bias and uncertainty

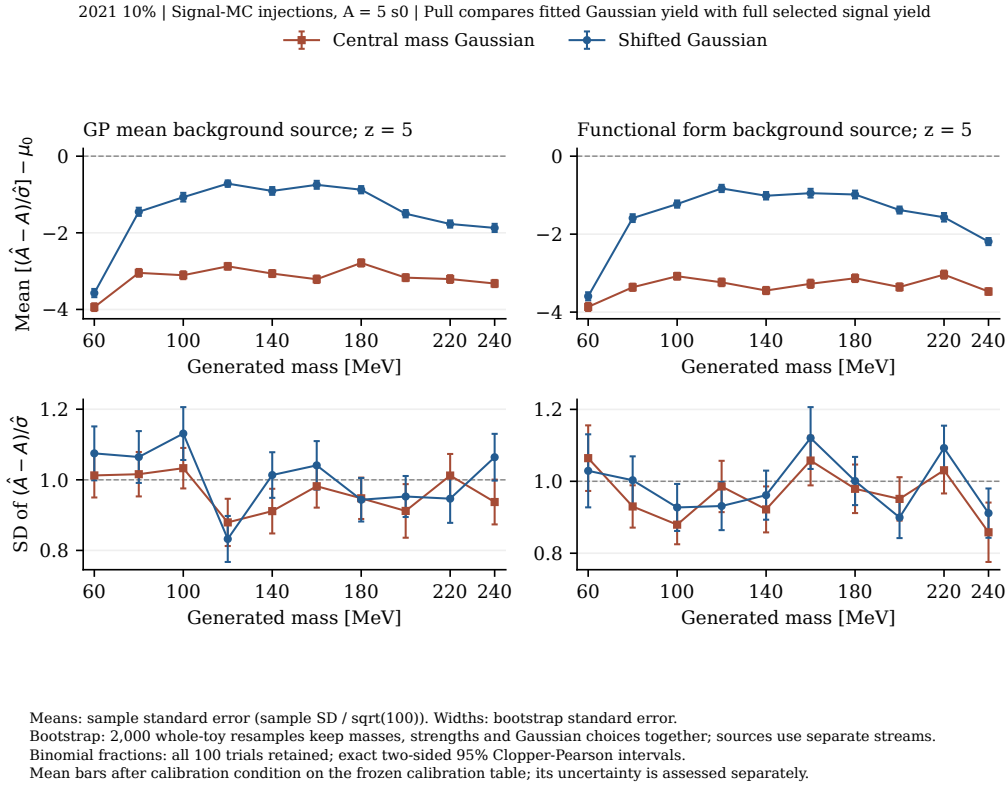


Figure 8. How to read the figure. Independent signal-MC evaluation experiments at $A = 5s_0$. The pull is $(\hat{A} - A)/\hat{\sigma}$: fitted yield minus the full selected injected yield, divided by the reported error. Top: its mean after subtracting the background-only mean pull μ_0 ; bars are sample standard errors. Bottom: its sample standard deviation; bars are bootstrap standard errors from 2,000 whole-experiment resamples. Each point uses 100 experiments. The ideal reference lines are zero for the mean and one for the width.

A negative mean pull says that the fitted Gaussian yield is below the injected full selected yield. A pull width near one only says that the fluctuation scale is roughly consistent with the reported error. It does not remove or excuse a displaced mean. This is why background-only centering can look satisfactory while the positive-signal test still shows a large bias.

The likelihood also provides a direct check. At the injected yield A , refit the background nuisance parameters and compute

$$q_A = 2\{\ell(\hat{A}, \hat{\theta}) - \ell(A, \hat{\theta}_A)\},$$

where ℓ is the log likelihood. The usual one-parameter Gaussian references accept $q_A \leq 1$ and $q_A \leq 3.841459$ for 68% and 95% likelihood sets. Here those sets are produced by a Gaussian-template model and checked against the full signal-MC yield. They need not contain that yield at the stated rates when the template response is biased.

The complete counts, always out of 100, and exact 95% binomial intervals remain in `results/heldout_summary.csv`. That inherited filename denotes independent evaluation. The next two sections distinguish an approximate corrected-yield diagnostic from the toy-calibrated confidence construction.

7. A calibrated yield and its error

The corrected point estimate subtracts the background-only yield bias and divides by the measured response:

$$A_c = \frac{\hat{A} - \delta}{R}, \quad R = \frac{\overline{\hat{A}_{s+b}} - \hat{A}_b}{3s_0} \text{ in the calibration cohort.}$$

For example, if the fitted yield is 600 events, the calibrated background-only bias is 100 events and $R = 0.5$, the corrected estimate is 1,000 selected events. Dividing by a small response amplifies the uncertainty as well as the yield.

Let $k_0 = \text{SD}\{(\hat{A}_0 - \delta)/\hat{\sigma}_0\}$ measure the background-only fluctuation scale. An approximate propagated variance is

$$\sigma_c^2 \simeq \frac{k_0^2 \hat{\sigma}^2 + \text{Var}(\delta) + A_c^2 \text{Var}(R) + 2A_c \text{Cov}(\delta, R)}{R^2}.$$

The bootstrap estimates δ and R together, retaining their covariance. The cross term has the displayed $+2A_c \text{Cov}(\delta, R)$ form; its numerical contribution can have either sign.

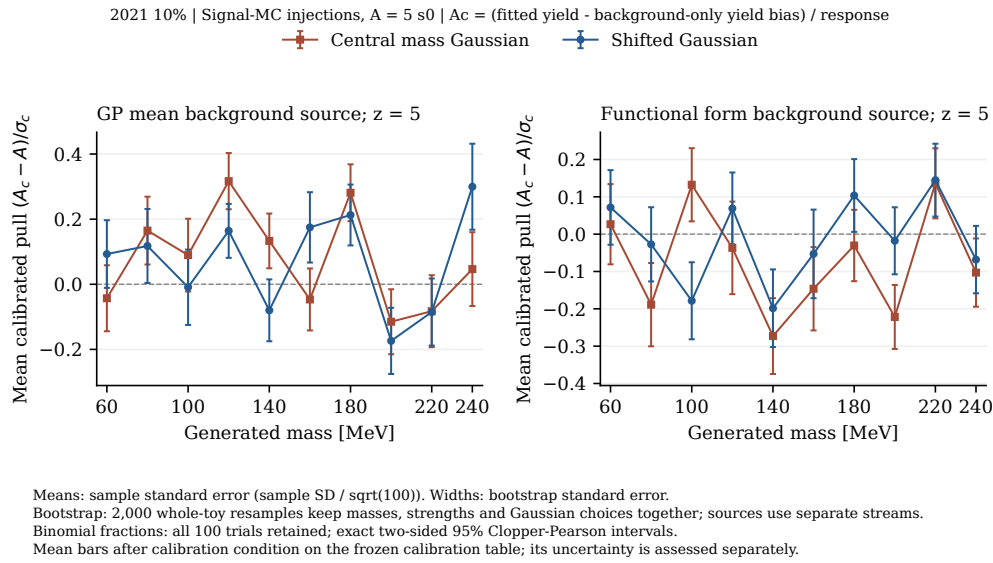


Figure 9. How to read the figure. Pull after yield calibration, $(A_c - A)/\sigma_c$, for independent signal-MC experiments at $A = 5s_0$. A mean near zero indicates agreement between calibrated and injected yields for this source and strength. Bars are sample standard errors across 100 evaluation experiments, conditional on the frozen calibration table. Each toy's σ_c includes the calibration-parameter terms in the equation above. The figure's bootstrap and sample-size definitions apply here.

This first-order uncertainty estimate treats the response as locally constant and uses the background-only width scale. It omits uncertainty in the pilot scale, the chosen background source and the finite signal-MC template. It is a diagnostic of a corrected estimator, not a new likelihood or an exact confidence interval. The primary limits below therefore calibrate the fitted statistic at every tested injected yield; they are not obtained by simply dividing an existing limit by R .

8. Turn the toy calibration into an upper-limit test

The key question is: *would an injected yield A commonly give a fit result this small?* For each preselected yield on the calibration grid, compare the observed (or evaluation) fitted yield with 100 calibration fitted yields generated at that same true A :

$$p_A = \frac{1 + \#\{\hat{A}_A^{\text{cal}} \leq \hat{A}^{\text{eval}}\}}{101}.$$

The numerator counts calibration results no larger than the result being tested, then adds one. Reject that yield when $p_A \leq 0.10$. A sufficiently small fitted yield is evidence against a large injected signal. This is a lower-side probability used for exclusion; Section 9 uses the opposite direction to ask about an excess above background.

Repeat this test at $z = 0, 1, 2, 3, 4, 5, 6, 8, 10, 12, 16, 20, 24$, with $A = zs_0$. Retain the *entire set* of accepted nodes. Its largest node is the displayed grid endpoint. We do not interpolate between simulated strengths. A missing interior node is recorded as a hole; acceptance of the largest node is recorded as right censoring, meaning that the grid does not reach the endpoint.

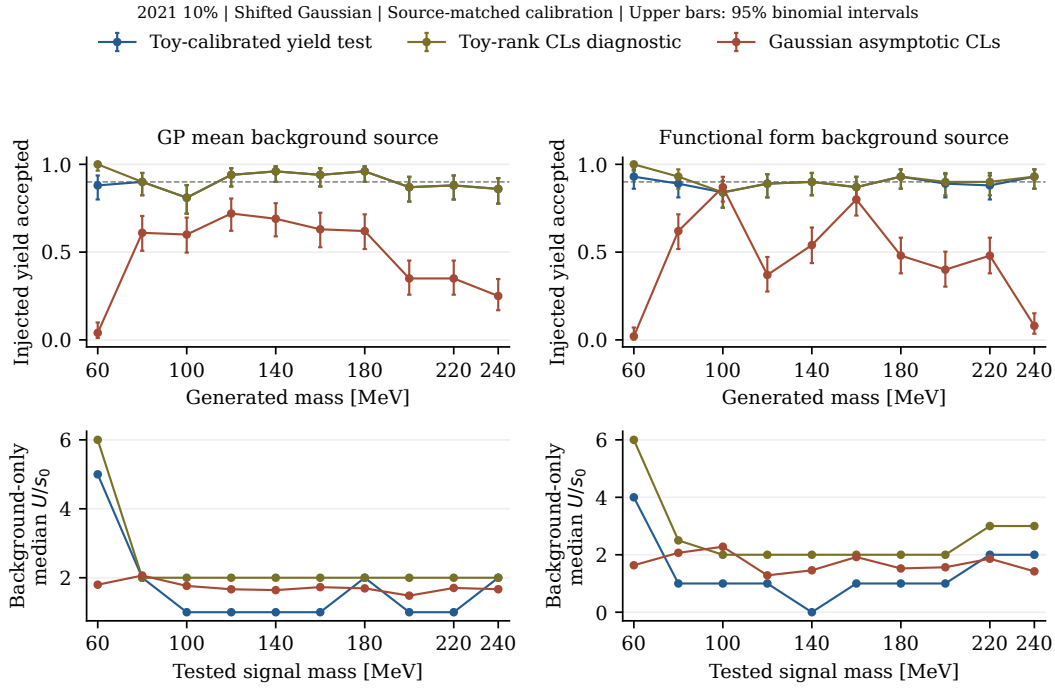
An empty set is unresolved, not a zero physical limit. If every node is rejected, the software stores endpoint zero together with an empty-set flag. If only zero is accepted, the grid resolves no behavior between zero and its first positive node. Neither case excludes all arbitrarily small positive signals. The accepted set and its flags must accompany the endpoint.

Why this is called calibration. The test uses distributions from signal-MC injections even though the fitted template is Gaussian. It therefore measures the distribution of the actual fitted statistic under the intended signal yield instead of assuming unit Gaussian-template response.

With continuous, exchangeable calibration and evaluation results, the add-one rank test rejects the true node in at most 10/101 of experiments, averaged over both calibration and evaluation cohorts. A single frozen table can have a different acceptance probability. The order-statistic reference gives a central 95% range of 83.60–95.10% for that probability. This calibration-table variation is separate from the binomial error of checking it with 100 evaluation experiments; ties make the stated rank rule conservative.

For comparison, the study also records a toy-rank CL_s diagnostic, $\min(1, p_A/p_{0,\text{lower}})$, and the inherited Gaussian-template asymptotic CL_s limit. The shared 1/101 rank floor can limit the toy-rank diagnostic's sensitivity. The Gaussian asymptotic reference uses profile-likelihood approximations and does not automatically describe the full signal-MC yield.

8. Check the procedure on independent experiments



Means: sample standard error (sample SD / $\sqrt{100}$). Widths: bootstrap standard error.
 Bootstrap: 2,000 whole-toy resamples keep masses, strengths and Gaussian choices together; sources use separate streams.
 Binomial fractions: all 100 trials retained; exact two-sided 95% Clopper-Pearson intervals.
 Mean bars after calibration condition on the frozen calibration table; its uncertainty is assessed separately.

Figure 10. How to read the figure. Shifted-Gaussian results, with calibration and evaluation using the same specified source. Upper panels: fraction of 100 independent experiments for which the injected $A = 5s_0$ is accepted (or is below the continuous Gaussian asymptotic endpoint). Bars are exact two-sided 95% Clopper-Pearson intervals. The dashed line marks 90%. Lower panels: median upper endpoint in background-only experiments, divided by the pilot yield-error scale s_0 ; no uncertainty bars are shown on these medians. Toy-based endpoints are grid maxima, while Gaussian asymptotic endpoints are continuous.

The toy-calibrated yield test retains the injected $A = 5s_0$ in 81–96% of GP-source evaluations and 84–93% of functional-source evaluations. The corresponding Gaussian asymptotic limits contain the full signal-MC yield in only 4–72% and 2–87%. This failure is why improving the core center alone is insufficient for full-yield inference.

Every planned evaluation ID remains in the tables; no fits failed in the main run. The denominator is always 100. The variation among masses includes finite evaluation statistics and fluctuations of the frozen calibration tables, so the plot should not be summarized as exact 90% coverage everywhere.

At zero injected yield, the stored numerical endpoint of an empty set also equals zero. Counting that endpoint as covering zero would hide the rejection of every tested node. The tables therefore retain both endpoint coverage and the separate, primary *truth-acceptance* field. Use the latter to assess the grid test.

9. What the observed 2021 spectrum shows

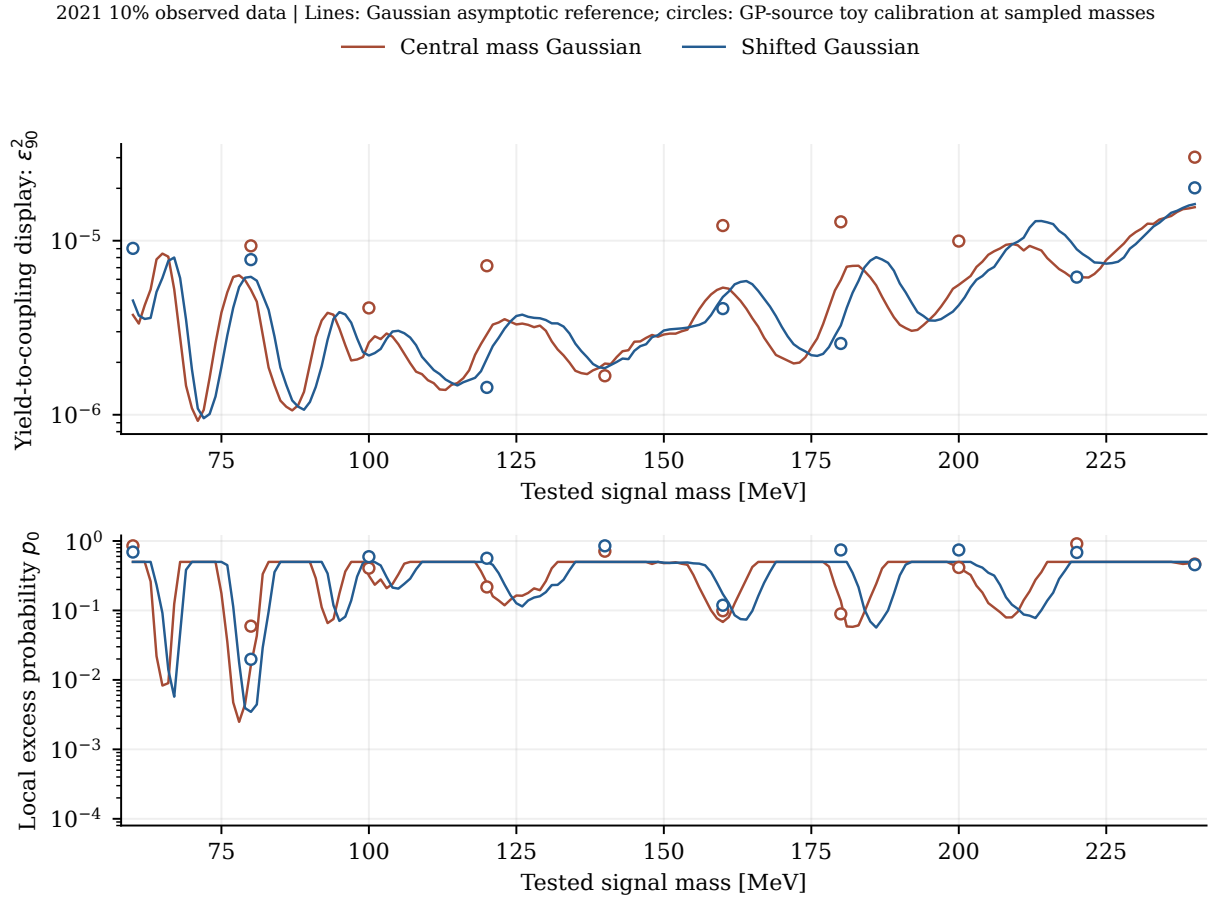


Figure 11. How to read the figure. 2021 10% observed data. Lines are Gaussian-template asymptotic results on a 1 MeV mass grid. Open circles are independent GP-source toy-calibrated results only at the simulated masses, 60–240 MeV in 20 MeV steps. Upper panel: yield limits converted to the inherited coupling display defined on the next page. Nonpositive grid endpoints are omitted from the logarithmic axis and retained with their flags in the tables. Lower panel: local excess probability. Lines use the profile-likelihood asymptotic approximation; open circles rank the observed fitted yield against background-only fitted yields. The line is not an interpolation of the points.

For the toy-based local excess probability, count how many of the 100 background-only calibration experiments return a fitted yield at least as large as the observed one:

$$p_0 = \frac{1 + k}{101}, \quad k = \#\{\hat{A}_0^{\text{cal}} \geq \hat{A}^{\text{obs}}\}.$$

This is the meaning of the former phrase “upper null tail.” The smallest resolvable rank is 1/101, even when there are no exceedances. A binomial interval for $k/100$ estimates the underlying background-only tail probability; it is distinct from the add-one rank.

The smallest local asymptotic p_0 moves from 0.00249 at 78 MeV for the central mass Gaussian to 0.00346 at 80 MeV for the shifted Gaussian. These masses were selected by scanning the data. Their local probabilities do not include the look-elsewhere effect and are not global discovery probabilities.

9. How fitted event yield is converted for display

The fit fundamentally measures an event-yield parameter. The displayed coupling variable uses the inherited conversion

$$A = \epsilon^2 C_{2021}(m), \quad C_{2021}(m) = \frac{3\pi m f_{\text{rad}}}{2\alpha} \rho(m).$$

Here $\alpha = 1/137$, f_{rad} is the archived effective radiative fraction, and $\rho(m)$ is the saved observed event density averaged around the **tested signal mass** m . The density average uses $m \pm 1.64\sigma_{\text{ref}}$, including fractional overlaps with histogram bins. Mass and resolution are in GeV, and the density is events per GeV.

The mass at which this conversion is evaluated does not move when the Gaussian center moves. The central and shifted Gaussian fits at a given tested mass use the same $C_{2021}(m)$. This specifies precisely how the event density enters the conversion. It separates a change in extraction shape from a change in the yield-to-coupling normalization.

The implementation fits $\psi = \epsilon^2/10^{-8}$, so the fitted event yield is $\hat{A} = \hat{\psi} C_{2021}(m) 10^{-8}$. The raw coupling columns are electron-channel proxies. For consistency with earlier visible-channel plots, the displayed values above the muon-pair threshold are multiplied once by

$$1 + \sqrt{1 - 4r}(1 + 2r), \quad r = (105.6583745 \text{ MeV}/m)^2, \quad m > 211.316749 \text{ MeV}.$$

Below threshold the factor is one. This display convention does not independently validate branching fractions, efficiency or selection equivalence. It is therefore not a detector-qualified physical exclusion.

How to interpret the two kinds of curve. The continuous asymptotic line answers a question within the fitted Gaussian likelihood approximation. The open points use signal-MC toy distributions and the finite accepted-yield grid. Their agreement or disagreement is informative, but neither a sparse set of calibrated points nor a dense asymptotic scan establishes a globally calibrated discovery probability.

Gaussian choice	Dense minimum mass [MeV]	Local asymptotic p_0
Central mass Gaussian	78	0.002488
Shifted Gaussian	80	0.003464

10. Combine campaigns while changing only 2021

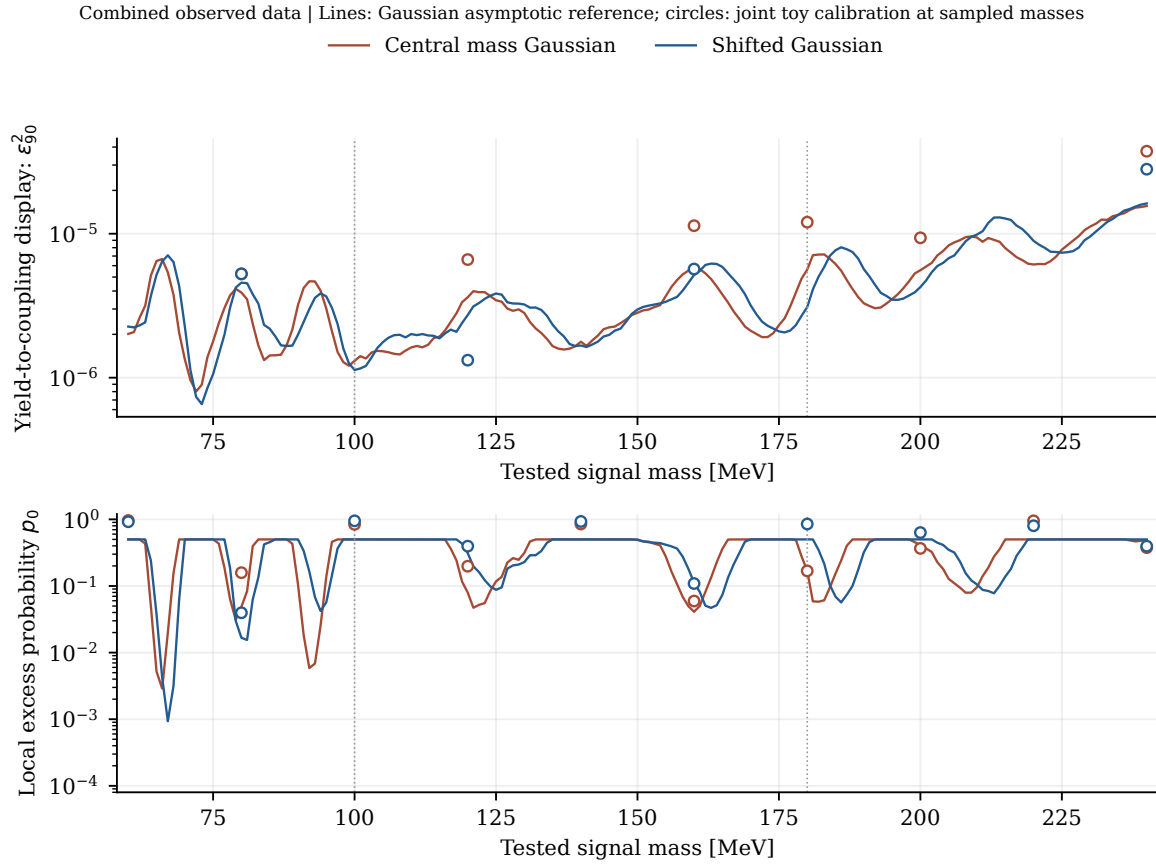


Figure 12. How to read the figure. Combined observed result with a common coupling parameter. The 2015 and 2016 models are unchanged; only the 2021 Gaussian center and its matched fit/training exclusion change. Lines are dense asymptotic references; open points are direct joint toy calibrations at simulated masses. All three campaigns contribute through 100 MeV, 2016+2021 through 180 MeV, and only 2021 above 180 MeV; dotted lines mark those boundaries. Empty and zero-only grid sets are omitted from the logarithmic endpoint panel. The coupling display is defined in Section 9.

The joint calibration simulates the full common-coupling experiment, rather than combining separately corrected amplitudes. It uses 100 pilot experiments to set the coupling scale, 100 independent GP-source calibration experiments at the thirteen strengths, and another 100 independent evaluation experiments at $z = 0, 1, 3, 5$. The 2021 signal is drawn from signal MC; older campaigns retain their inherited signal models.

For the combined *excess* probability, the ranked statistic is the signed profile-likelihood root. It is the square root of twice the best-fit log-likelihood improvement over background only, with the sign of the fitted coupling. Thus the background-only exceedance count uses likelihood evidence for an excess, not the standalone 2021 fitted yield. Exclusion still orders the fitted yield parameter.

The smallest combined local asymptotic p_0 changes from 0.00289 at 66 MeV to 0.000933 at 67 MeV. These mass-selected minima remain descriptive local references. Changing only the 2021 model does not independently recalibrate the older campaigns.

11. Check the combined toy calibration

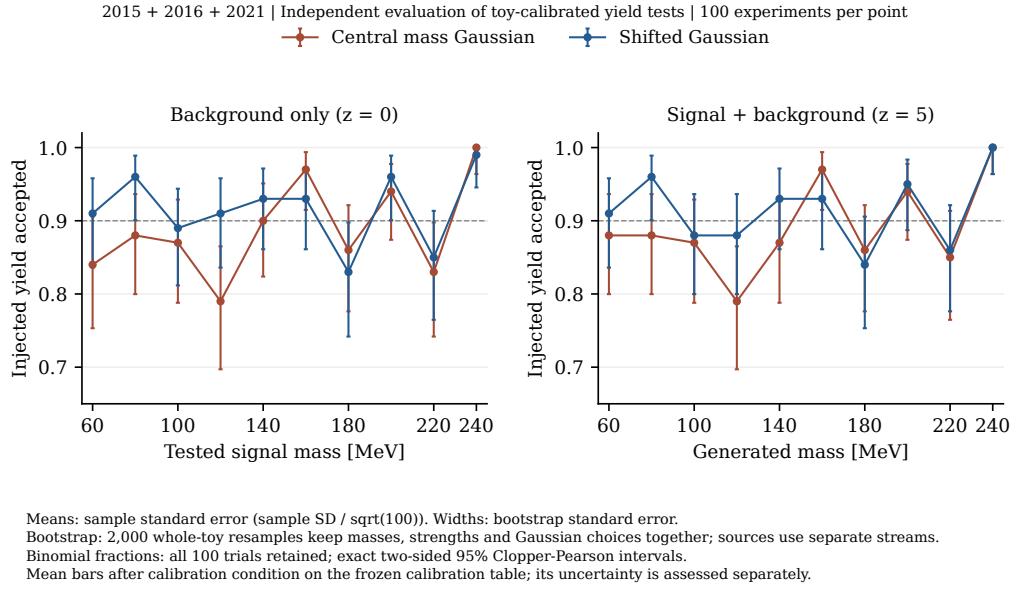


Figure 13. How to read the figure. Fraction of independent joint evaluation experiments retaining the injected coupling at its grid node. Left: background only; right: signal plus background at $z = 5$ in the joint pilot-defined coupling unit. Each point uses 100 experiments and exact two-sided 95% Clopper–Pearson intervals. The two Gaussian choices share each whole experiment, preserving their correlation. Calibration and evaluation use the GP mean background source; campaign support follows Section 10.

Across all joint evaluation cells, acceptance ranges from 78% to 100%; 260 of 8,000 accepted-node sets are empty. This reflects a finite, frozen calibration with discrete strengths. It does not establish exactly 90% coverage for each mass and strength.

The observed joint inversions have five empty sets and five additional zero-only sets among twenty mass/fit-choice pairs. None accepts the largest grid node or has an internal hole. Their zero endpoints are unresolved continuous limits, not exclusions of arbitrarily small signals.

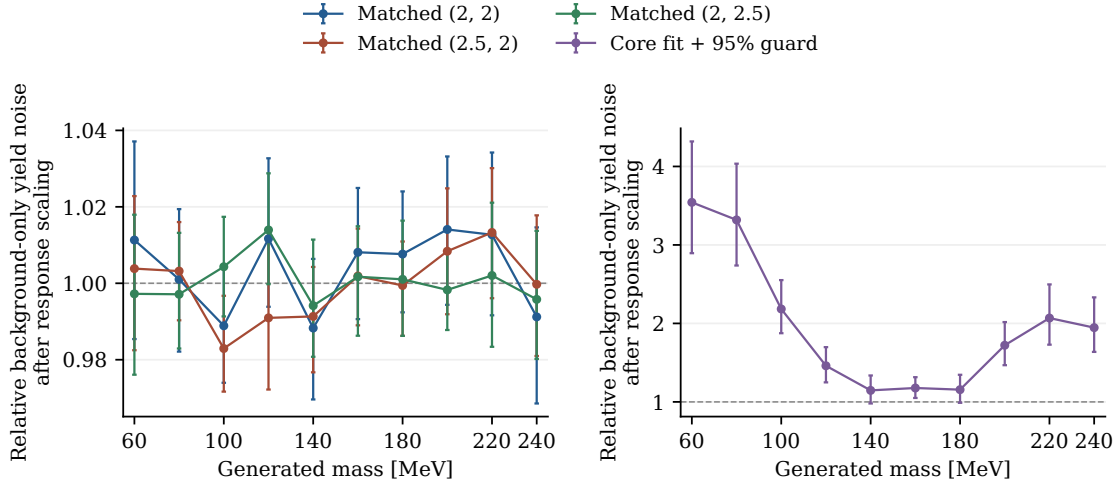
Mass [MeV]	Central p_0	Shifted p_0	Central grid result	Shifted grid result
60	0.9604	0.9208	empty	empty
80	0.1584	0.0396	5.26e-06	5.26e-06
100	0.8416	0.9505	zero only	empty
120	0.1980	0.3960	6.62e-06	1.32e-06
140	0.8515	0.9307	zero only	empty
160	0.0594	0.1089	1.14e-05	5.69e-06
180	0.1683	0.8515	1.20e-05	zero only
200	0.3663	0.6337	9.37e-06	zero only
220	0.9505	0.8020	empty	zero only
240	0.3762	0.3960	3.74e-05	2.81e-05

Positive grid entries are endpoints in the coupling display of Section 9. The smallest joint empirical rank is $4/101 = 0.03960$ at 80 MeV for the shifted Gaussian, corresponding to three background-only exceedances. This finite sample does not empirically resolve the much smaller dense asymptotic minimum.

12. Would a different fit or training window help?

A wider excluded training region removes more signal contamination from the background estimate, but also removes background information. The relevant test is the uncertainty on the full signal yield after accounting for response, not signal containment alone.

The baseline fit and excluded training region both extend $2.25\sigma_{\text{ref}}$ on each side of the shifted center. Three alternatives change both together to left/right half-widths (2, 2), (2.5, 2) and (2, 2.5) in units of σ_{ref} . A fourth keeps the baseline fit window and widens only the excluded training region to contain the central equal-tail 95% of selected signal MC.



2021 10% | Signal MC at $A = 3s_0$ on the GP mean background source | 100 evaluation toys
 Bars: 95% percentile intervals from 2,000 whole-toy bootstrap resamples.
 Resampling keeps masses, signal strengths and window choices together; pilot scale is fixed.
 Ratio = $[\text{SD}(\hat{A}_0)/R] / [\text{signal response}] / \text{baseline value}$. Lower is better.

Figure 14. How to read the figure. Separate GP-source side study with 100 pilot and 100 independent evaluation backgrounds. Signal-MC injections use a common $A = 3s_0$ and the same experiments across window choices. The plotted ratio compares $\text{SD}(\hat{A}_0)/R$ for each choice with the baseline: a lower value means less full-yield noise. Bars are 95% percentile intervals from 2,000 whole-toy bootstrap resamples, preserving mass and window-choice correlations; the pilot scale is fixed. These are sensitivity diagnostics, not calibrated upper limits.

The three matched-window alternatives give ratios 0.983–1.014, mostly with intervals spanning one. The wider-left 100 MeV case is 0.983 [0.972, 0.997], one exploratory point among thirty comparisons. This does not establish a reliable improvement. The 95% training exclusion worsens the noise by factors 1.15–3.54, despite reducing training leakage to about 5%.

At 60 MeV that equal-tail exclusion spans roughly 51.0–212.25 MeV, leaving a large gap for background prediction. Keeping three external bins on either side checks geometry, not the reliability of that extrapolation. The baseline $\pm 2.25\sigma_{\text{ref}}$ prescription is retained.

The shape coordinate $u = (m_{\text{rec}} - c)/\sigma_{\text{core}}$ from Section 1 must not be confused with these widths in σ_{ref} . Translating a boundary between the two requires the ratio $\sigma_{\text{core}}/\sigma_{\text{ref}}$.

13. What is established, and how to reproduce it

Established within this study. A shifted Gaussian improves extraction of the supplied signal MC. Background-only bias and incomplete signal response are different effects and can be measured separately. A frozen toy-calibrated yield test gives a defined conditional inference procedure that can be checked on independent experiments. The chosen baseline window remains appropriate among the tested alternatives.

Scope of that conclusion. The background sources, empirical signal-MC histogram and archived GP kernel settings are fixed. The selected TC signal-MC metadata identifies v13 inputs; it does not establish full v13/v16 selection equivalence or daughter association. Uncertainty in the finite signal-MC template, other possible background shapes, model selection and a global look-elsewhere correction are outside this ensemble. Earlier 2016 support/optimizer qualifications remain unresolved by these 2021 toys. A residual in one observed spectrum is not itself an ensemble bias estimate.

Numerical evidence. All main planned fits, truth profiles and Gaussian asymptotic endpoints passed the archived finite/positive, score and Hessian checks. The original validation includes seed replay, probability normalization, paired draws, frozen-calibration hashes and representative direct calculations. The main-body revision reruns no inference fits. The appendices add new v16 signal-MC core fits. Hash comparisons verify that the numerical inputs, cohorts, calibration and result tables are unchanged from the supplied v6.3.5 archive.

Reproduction. The package contains `source/report.tex`, the plot and narrative builders in `scripts/`, figure source data, and the original numerical checkpoints. The README gives the report-only rebuild commands. `protocol.json` deliberately retains the v6.3.5 scientific protocol and seed so that editorial versioning cannot be mistaken for a new experiment. `revision.json` describes v6.3.6. The source image is preserved unchanged with a SHA-256 digest. New and inherited QA records are identified separately.

The source keys `nominal`, `functional`, `pole` and `logshift` are preserved in machine-readable files for compatibility. In this note they mean GP mean background source, functional form background source, central mass Gaussian and shifted Gaussian, respectively. Inherited “heldout” filenames contain independent evaluation results. No terminology change alters the estimator or data.

Related studies. The v6.1 signal-MC study supplies the catalogue, core fits and logarithmic center law. The v6.2 and v6.3.1 matched-template studies generated and fitted signal MC; their closure does not by itself validate a Gaussian fit to signal-MC injections. The v6.3.2 study concerned controlled 2016 exposure changes. The main-body inference results are the frozen v6.3.5 results. The appendices provide the new v16 TC and UC signal-shape studies.

Statistical references retained from v6.3.5. The PDG statistics review discusses bias, confidence constructions and CL_s : <https://pdg.lbl.gov/2026/reviews/rpp2026-rev-statistics.pdf>. The profile-likelihood asymptotic reference is Cowan et al., <https://arxiv.org/abs/1007.1727>. The finite-rank rule actually used here is given explicitly in Section 8.

Appendix A. Locate and verify the v16 signal samples

The nearby v16 production contains both requested signal samples. The target-constrained (TC) and unconstrained (UC) categories are established by the stored vertex-type branch, not inferred from their directory names. Both are selected 2021 signal Monte Carlo; this appendix studies their reconstructed mass shapes without an observed-data fit.

The common directory is `/sdf/data/hps/physics2021/preselection/v16/`. Its `ap_signal_prompt_smeared` child contains TC vertices, while `ap_signal_prompt_dis_smeared` contains UC vertices. We read the latest `preselection` tree once per file, using `vertex./vertex.invM_` in GeV. No extra cuts, smearing or recentering are applied to the stored masses.

Verified quantity	TC	UC
Generated samples	60–260 MeV, step 20	60–260 MeV, step 20
Source ROOT files	24	24
Selected candidate rows	17,775,543	21,089,537
Nonunit event weights	0	0
Underflow / overflow	0 / 1,205	0 / 1,481
Stored vertex type	TargetConstrained	Unconstrained

Histograms use 0.1 MeV bins on 0–400 MeV. Their bin sums plus flow counts reproduce all selected rows. Generated truth masses agree with the sample labels within 0.003914 MeV. All 22 samples have finite reconstructed masses and positive unit weights. As in the main note, 260 MeV is a shape control; the shift relations and common-core comparisons use 60–240 MeV. There is no independent 65 MeV sample.

These are different selected productions. TC cutflow labels list electron/positron two-dimensional-hit thresholds of 8/10 and momentum sum above 2.8 GeV; UC labels list 6/8 and momentum sum above zero. Both display a vertex- $\chi^2 < 30$ step, but the TC count does not decrease there in any file, whereas the UC count does. The timing labels are 5.7 and 9.2 ns in both productions, so a v16 directory name alone does not certify equivalence to the observed-data selection.

The stored momentum-smear ratios equal one in TC and are generally nonunit in UC. These auxiliary branches do not by themselves establish the transformations already included in the stored mass. We therefore histogram that mass exactly once. A TC–UC shift difference can reflect vertex treatment, selection and smearing conventions together; it is not an isolated measurement of the effect of the vertex constraint.

Reproducible extraction. A bounded read-only S3DF pass streamed the mass and audit branches with one decompression worker. The saved provenance contains all file paths, sizes, modification times, tree cycles, cutflows and SHA-256 hashes of streamed branch payloads. These hashes identify the analyzed payloads, not the complete ROOT files. All core fits and plots were then made locally. The main inference results and their frozen inputs remain unchanged; no UC upper limit is calculated because a matching observed UC source histogram is not available.

Appendix A. Use the same core definition for both samples

The goal is to measure the reconstructed signal core without forcing its asymmetric tail into a single Gaussian. The same algorithm used for v6.1 is applied separately to every v16 TC and UC histogram.

First smooth the histogram only to locate its peak within three reference widths of the generated mass. Then fit the original, unsmoothed bin counts within $1.5\sigma_{\text{ref}}$ of that peak with a bin-integrated Gaussian plus a nonnegative linear pedestal. The pedestal describes the local signal tail under the core; it is not an experimental background. The Gaussian center c_{core} and width σ_{core} are free. The mass displacement is $c_{\text{core}} - m$.

The TC reference-resolution formula from Section 1 supplies the search and fit-span scale for both categories. In UC it is only a common numerical scale used to define the fit region, not an assumed or calibrated UC detector resolution. The UC Gaussian width is fitted freely. As a stability check, repeat the fit with half-spans $1.25\sigma_{\text{ref}}$ and $2\sigma_{\text{ref}}$.

Uncertainty on a core position. Each mass histogram is independently resampled 32 times by drawing a Poisson count in every bin with mean equal to that bin's observed signal-MC count. The error bar is the sample standard deviation of the 32 refitted centers. All 22 central fits, all 66 fit-span checks and all 704 resampling fits passed the optimizer, boundary and peak-location checks. These bars describe finite signal-MC statistics conditional on the selected production and core definition. They are not standard errors of 100 background toys, and they do not include selection, smearing or fit-definition uncertainty.

Which shift relation describes the mass dependence? Fit four simple functions by unweighted least squares to the ten core displacements from 60 to 240 MeV:

$$a, \quad a + bx, \quad a + b \ln[m/(150 \text{ MeV})], \quad a + bx + dx^2, \quad x = \frac{m - 150 \text{ MeV}}{100 \text{ MeV}}.$$

To compare them, omit one mass, fit the remaining nine, and predict the omitted displacement. Repeat for all ten masses and report the root-mean-square (RMS) prediction error in MeV. The preferred description has the smallest error among these four choices. This is a model-selection diagnostic, not independent validation or a guarantee outside the sampled domain. All coefficients describe $c - m$ in MeV.

Can one shape describe the cores? Align each histogram using its own fitted c_{core} and σ_{core} . A core-only comparison normalizes each distribution within $|u| < 2$, where $u = (m_{\text{rec}} - c_{\text{core}})/\sigma_{\text{core}}$. A second panel retains full selected normalization to expose the tails. The common empirical curve gives equal weight to each generated mass; it is not a Gaussian fit. The quantitative comparison is the largest difference of normalized cumulative probabilities inside the core, with the tested mass omitted from the common curve.

Appendix B. The v16 target-constrained (TC) core shift

The v16 TC sample reproduces the earlier logarithmic core-shift pattern. The logarithmic relation gives the smallest omitted-mass prediction error of the four tested forms:

$$c_{\text{TC}}(m) - m = [-3.224325 - 2.213983 \ln[m/(150 \text{ MeV})]] \text{ MeV}.$$

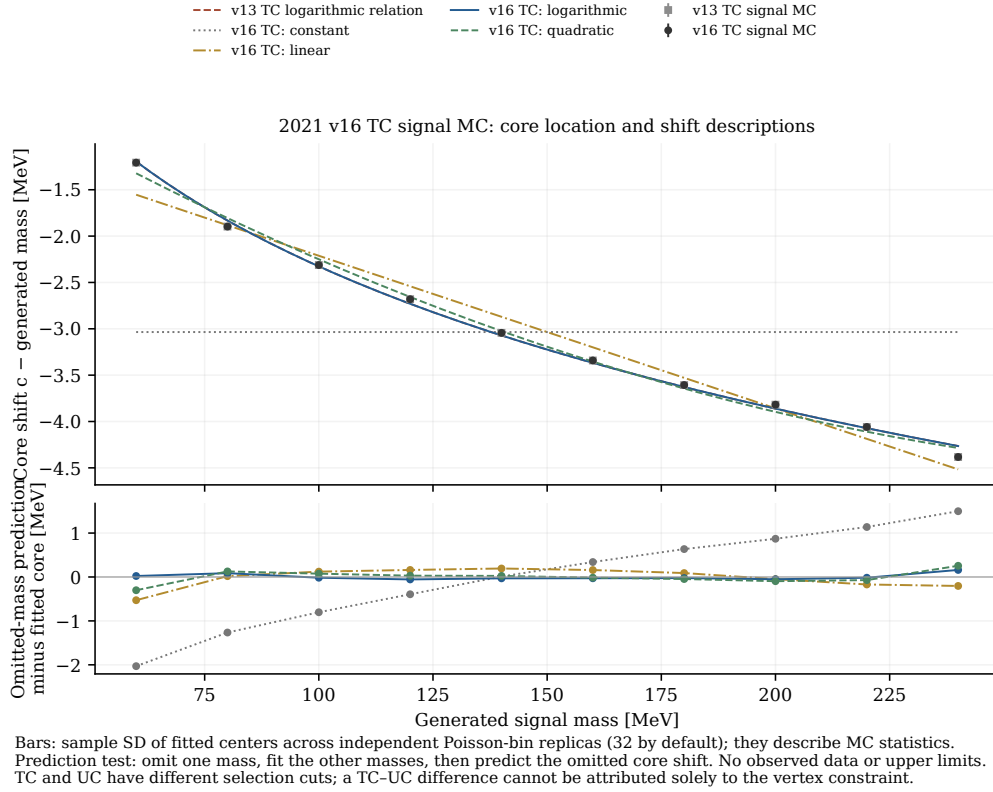


Figure 15. Upper panel: v16 TC fitted core displacement and four fitted descriptions; the original v13 TC relation supplies context. Lower panel: predicted minus measured displacement when each mass is omitted from its relation fit. Points use generated masses 60–240 MeV. Error bars on fitted centers are sample standard deviations from 32 independent Poisson-bin resamples per histogram; the prediction-error curves have no uncertainty bars. These descriptive shape comparisons do not establish sample independence or full selection equivalence. No significance is assigned to differences between productions.

Shift description	Omitted-mass RMS [MeV]
Constant	1.0647
Linear	0.2153
Logarithmic	0.0652
Quadratic	0.1399

Its omitted-mass RMS error is 0.0652 MeV. At the directly simulated masses, the largest v16–v13 TC core-center difference is only 0.000078 MeV. The histograms are very close but not identical. This comparison does not establish independent samples or full selection equivalence. The core-definition span check moves a center by up to 0.162 MeV, a separate effect from the statistical bars.

Appendix B. The TC signal-MC catalogue

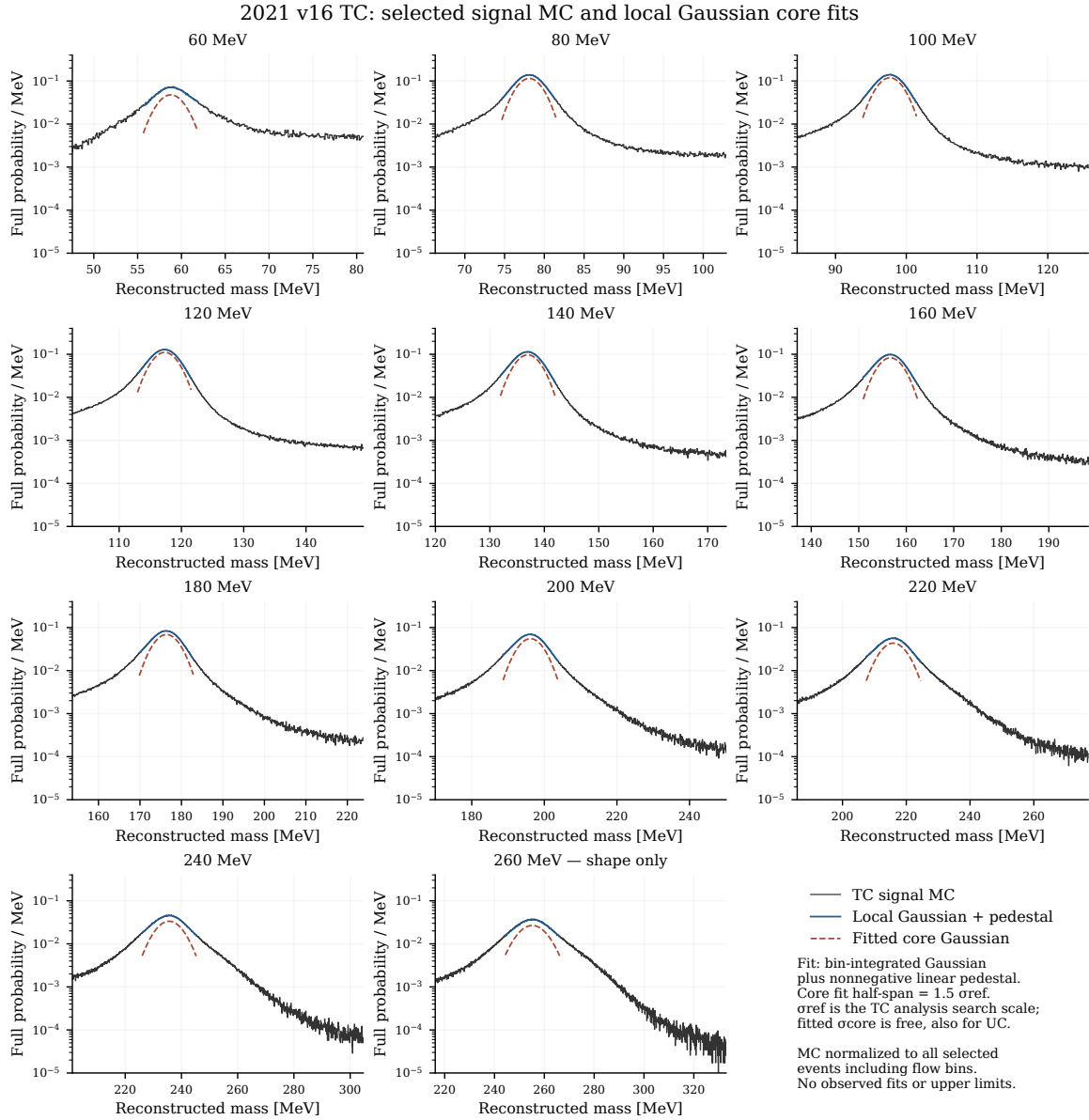
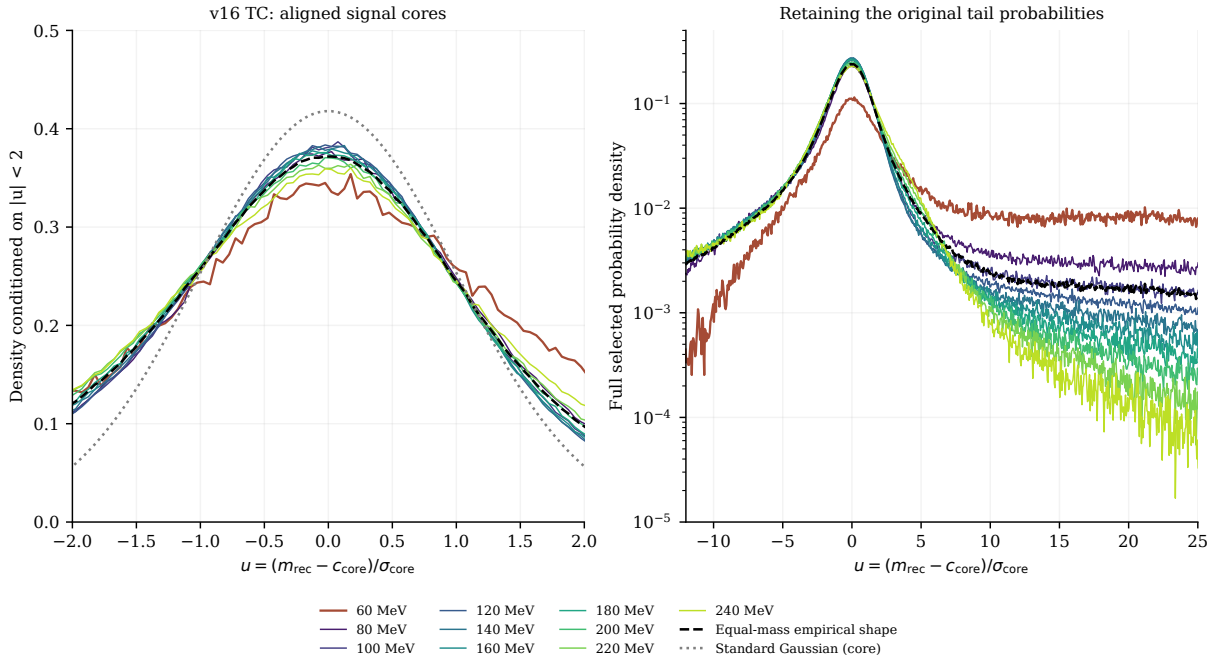


Figure 16. All eleven v16 TC samples. Black: selected signal-MC density, normalized to all selected candidates including overflow. Blue: local Gaussian plus nonnegative linear pedestal, drawn only across its fitted interval. Red dashed: the Gaussian component of that local fit, with its fitted amplitude and free width. These curves summarize the core; they are not the unit-area extraction Gaussians used in the main inference study. The logarithmic vertical scale shows the tails. The 260 MeV panel is shape-only and is excluded from the fitted mass-shift relations. No additional smearing or mass translation was applied.

A good description of the local peak does not mean that the fitted Gaussian contains the full selected signal probability. The pedestal is a local fitting device; it is not extrapolated as a physical full-range signal model. The next page compares aligned cores while retaining the original tails in a separate panel.

Appendix B. How similar are the aligned TC cores?



Centers and widths come from each sample's own local fit. Comparison of core shape conditions on those fitted values. Empirical lines, no error bars; 260 MeV is excluded from common-shape and shift-law fits. No efficiencies or upper limits.

Figure 17. v16 TC distributions aligned by each sample's fitted core center and width. Left: each curve is normalized within $|u| < 2$; right: full selected normalization retains the original tail probabilities. Black dashed: equal-mass empirical common shape. Gray dotted: standard Gaussian conditioned on $|u| < 2$. Individual curves are signal-MC histograms; no error bars are shown. The core-only normalization is used solely for this shape comparison, never to renormalize an injected signal yield. The comparison uses 60–240 MeV.

From 80 to 240 MeV, the largest omitted-mass cumulative-probability difference inside the core ranges from 0.0067 to 0.0137. This supports a common empirical core description at the percent level in this specific diagnostic. It does not establish a universal full signal shape, common acceptance or an exact Gaussian core.

At 60 MeV the corresponding difference is 0.0450, and only 32.39% of the full selected probability lies within two fitted core widths. The 80–240 MeV core fractions range from 60.73% to 71.46%. Thus the low-mass sample still needs separate treatment even after its center and width are aligned. The near identity of the v13 and v16 TC core law supports the stability of the center prescription in the main note. It does not replace the original injection calibration, assess all selection changes, or propagate finite signal-template uncertainty into those limits. The existing TC inference results are retained unchanged. This appendix evaluates the shift and shape evidence rather than rerunning observed limits or the full injection calibration. **Files.** The extracted histograms and per-file audit metadata are in `appendix/inputs/`; core fits, all resampling fits, alternative spans, prediction checks and shape metrics are in `appendix/results/`. The analysis and figure source is `appendix/scripts/analyze_shifts.py`. These data are sufficient to rebuild the appendix locally without another S3DF read.

Appendix C. The v16 unconstrained (UC) core shift

UC also reconstructs below the generated mass, but its displacement is smaller and has a different mass dependence. Among the four tested forms, the quadratic gives the smallest omitted-mass prediction error:

$$c_{\text{UC}}(m) - m = \left[-0.974838 - 0.791808x - 0.296125x^2 \right] \text{MeV}, \quad x = \frac{m - 150 \text{ MeV}}{100 \text{ MeV}}.$$

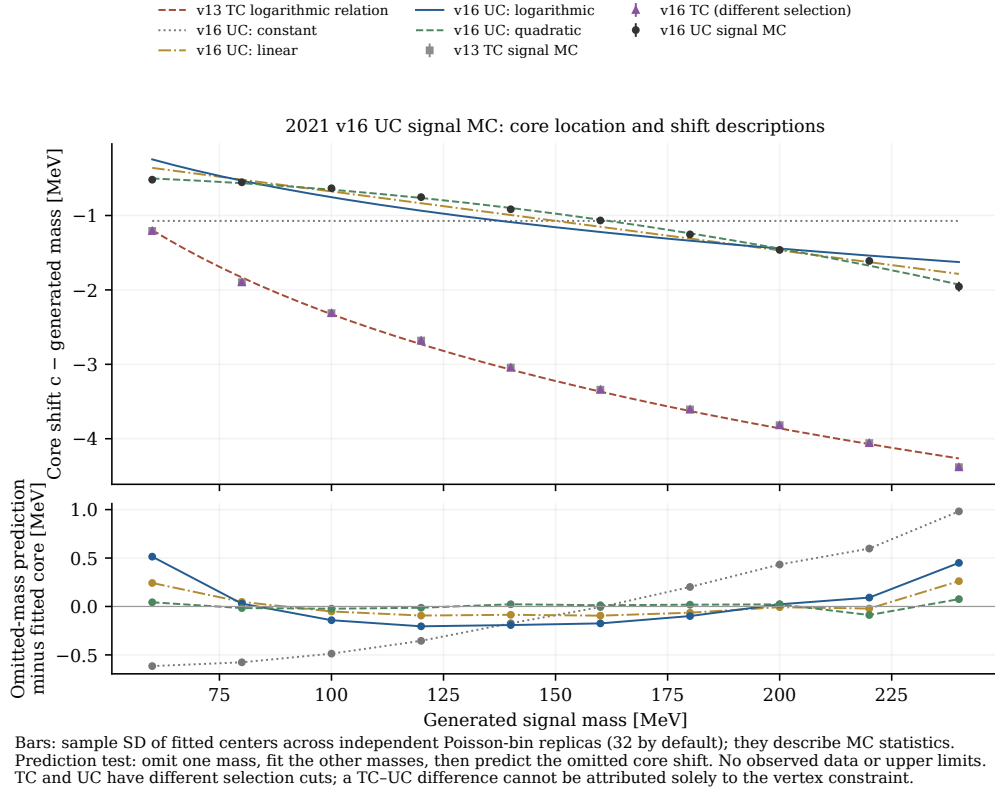


Figure 18. Upper panel: v16 UC fitted core displacement and four fitted descriptions; the original v13 TC relation supplies context. Lower panel: predicted minus measured displacement when each mass is omitted from its relation fit. Points use generated masses 60–240 MeV. Error bars on fitted centers are sample standard deviations from 32 independent Poisson-bin resamples per histogram; the prediction-error curves have no uncertainty bars. These descriptive shape comparisons do not establish sample independence or full selection equivalence. No significance is assigned to differences between productions.

Shift description	Omitted-mass RMS [MeV]
Constant	0.5152
Linear	0.1271
Logarithmic	0.2482
Quadratic	0.0422

The UC displacement runs from -0.519 MeV at 60 MeV to -1.956 MeV at 240 MeV. The quadratic prediction RMS is 0.0422 MeV, compared with 0.2482 MeV for a refitted logarithmic form. Thus the TC logarithmic correction should not be copied into UC. Changing the core-fit span moves a center by up to 0.088 MeV in this domain; the statistical bars do not include that variation.

Appendix C. The UC signal-MC catalogue

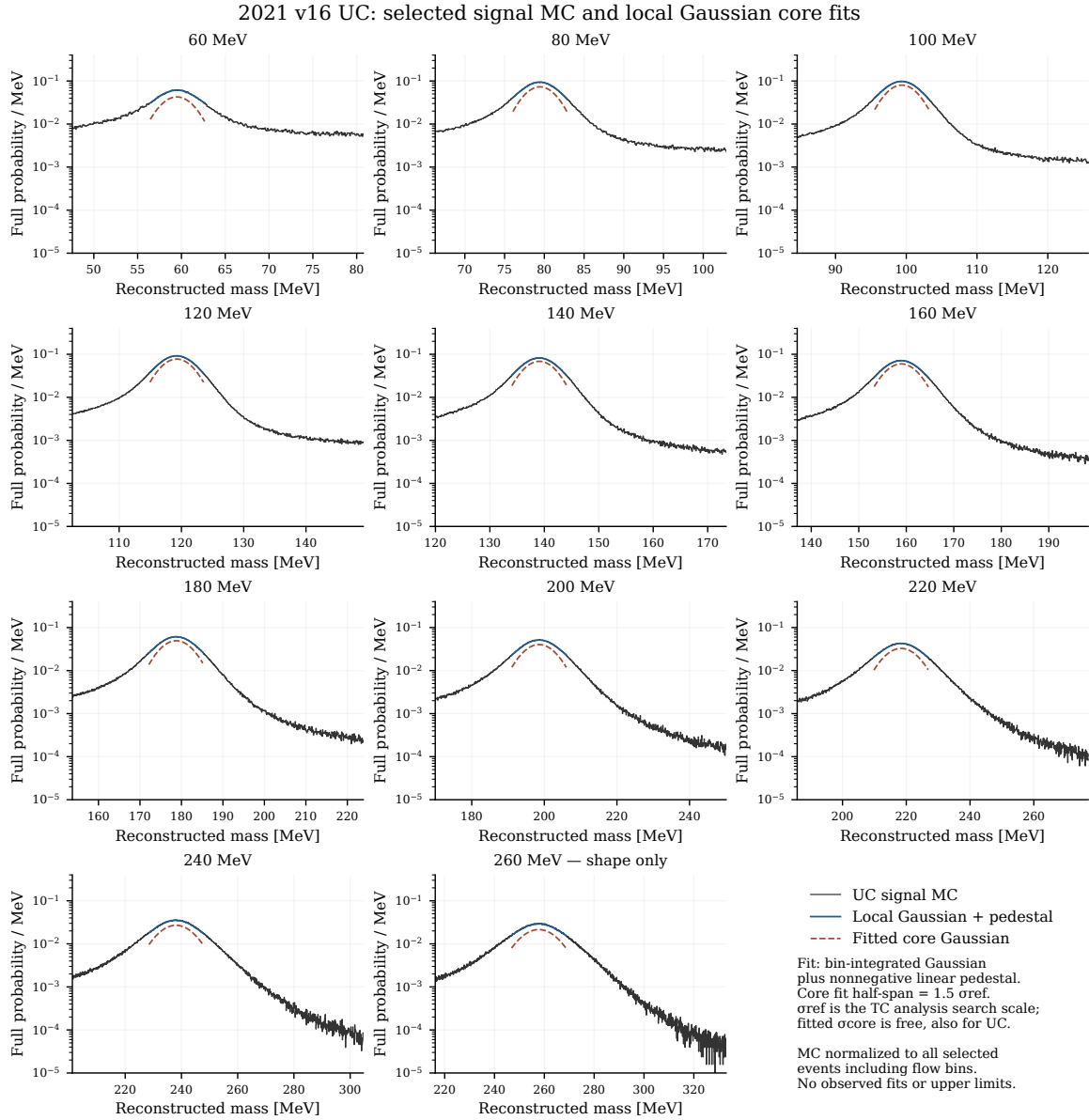
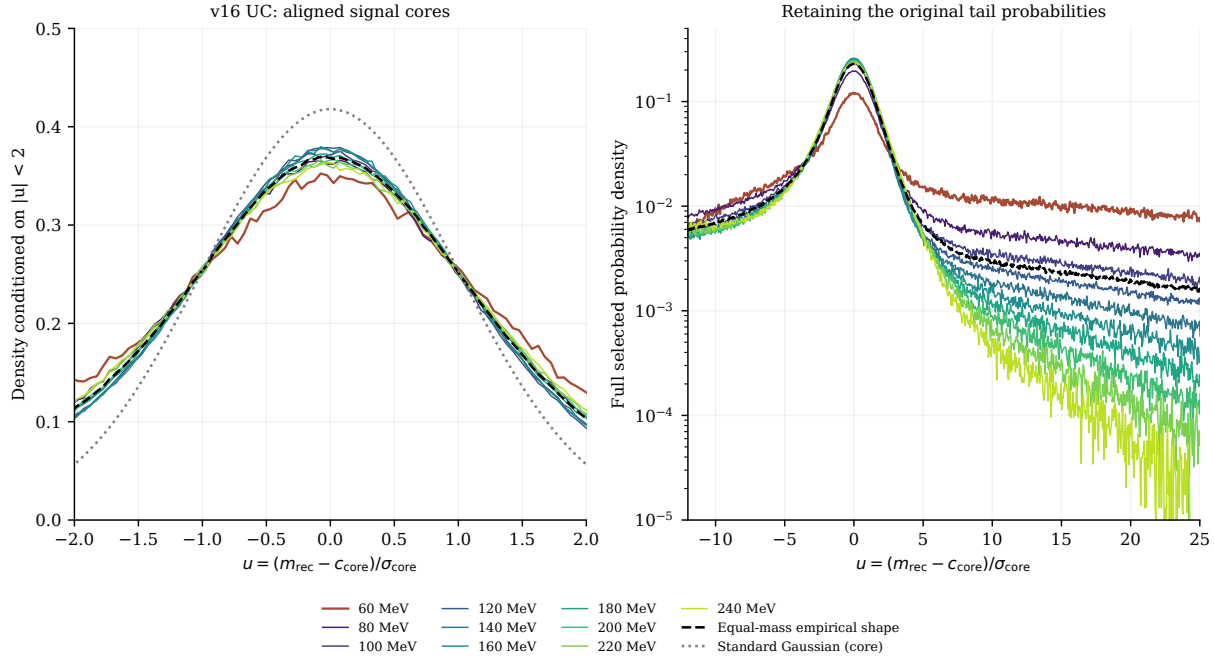


Figure 19. All eleven v16 UC samples. Black: selected signal-MC density, normalized to all selected candidates including overflow. Blue: local Gaussian plus nonnegative linear pedestal, drawn only across its fitted interval. Red dashed: the Gaussian component of that local fit, with its fitted amplitude and free width. These curves summarize the core; they are not the unit-area extraction Gaussians used in the main inference study. The logarithmic vertical scale shows the tails. The 260 MeV panel is shape-only and is excluded from the fitted mass-shift relations. No additional smearing or mass translation was applied.

A good description of the local peak does not mean that the fitted Gaussian contains the full selected signal probability. The pedestal is a local fitting device; it is not extrapolated as a physical full-range signal model. The next page compares aligned cores while retaining the original tails in a separate panel.

Appendix C. How similar are the aligned UC cores?



Centers and widths come from each sample's own local fit. Comparison of core shape conditions on those fitted values. Empirical lines, no error bars; 260 MeV is excluded from common-shape and shift-law fits. No efficiencies or upper limits.

Figure 20. v16 UC distributions aligned by each sample's fitted core center and width. Left: each curve is normalized within $|u| < 2$; right: full selected normalization retains the original tail probabilities. Black dashed: equal-mass empirical common shape. Gray dotted: standard Gaussian conditioned on $|u| < 2$. Individual curves are signal-MC histograms; no error bars are shown. The core-only normalization is used solely for this shape comparison, never to renormalize an injected signal yield. The comparison uses 60–240 MeV.

From 80 to 240 MeV, the largest omitted-mass cumulative-probability difference inside the core ranges from 0.0026 to 0.0079. This supports a common empirical core description at the percent level in this specific diagnostic. It does not establish a universal full signal shape, common acceptance or an exact Gaussian core.

At 60 MeV the corresponding difference is 0.0180, and only 34.48% of the full selected probability lies within two fitted core widths. The 80–240 MeV core fractions range from 53.63% to 68.74%. Thus the low-mass sample still needs separate treatment even after its center and width are aligned. The UC core is broader than the TC core in these samples, and its smaller shift has a different mass dependence. These conclusions concern the complete selected productions, whose selection and smearing information differ. The agreement in sign with the HPS Internal comparison in Section 1 is qualitative; that supplied image does not provide enough provenance for a quantitative combined fit.

No UC upper limit, local excess probability or signal-recovery calibration is computed. Those would require the matching observed UC histogram and an appropriate background and yield model. The appendix supplies the signal-shape evidence needed to design that later study. **Files.** The extracted histograms and per-file audit metadata are in `appendix/inputs/`; core fits, all resampling fits, alternative spans, prediction checks and shape metrics are in `appendix/results/`. The analysis and figure source is `appendix/scripts/analyze_shifts.py`. These data are sufficient to rebuild the appendix locally without another S3DF read.